ANGLO-CHINESE JUNIOR COLLEGE
DEPARTMENT OF CHEMISTRY
Promotional Examination

CHEMISTRY **9729**
Higher 2

Question Booklet 06 October 2020
2 hours

Additional Materials: Multiple Choice Answer Sheet
Data Booklet
Answer Booklet

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Write in dark blue or black pen **except** on the Multiple Choice Answer Sheet.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluids.

A Data Booklet is provided.
You are reminded of the need for good English and clear presentation in your answers.
The use of an approved scientific calculator is expected, where appropriate.

Sections A, B and C
Answer **all** questions.

Section A
Write in soft pencil.
There are **twenty-five** questions in Section A. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Sections B and C
Questions for Sections B and C are found in the Answer booklet. Answer them on the Answer Booklet.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Section A: Multiple Choice Questions (25 marks)

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct and indicate your answers on the separate answer sheet provided.

1 An unknown compound **P** containing only carbon, hydrogen and oxygen was subjected to combustion analysis. 0.1 g of **P** on complete combustion gave 0.228 g of CO_2 and 0.0931 g of H_2O.

What is the molecular formula of **P**?

A C_2H_3O **B** C_3H_6O **C** C_3H_8O **D** $C_2H_6O_2$

2 12.50 cm^3 of a 0.800 mol dm^{-3} solution of a metallic salt is oxidised exactly by 40.00 cm^3 of 0.250 mol dm^{-3} acidified $H_2O_2(aq)$.

If the original oxidation number of the metal in the salt was +2, what would be the new oxidation number?

A +3

B +4

C +5

D +6

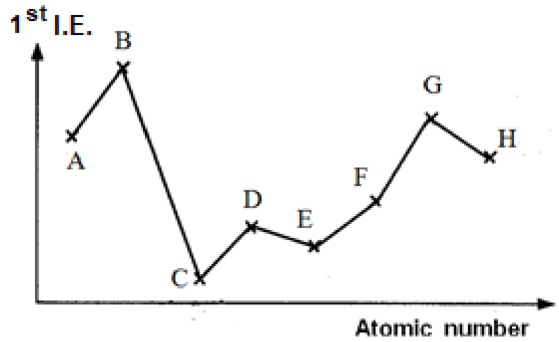
3 The successive ionisation energies (IE) of two elements, **Q** and **R**, are given below.

IE / kJ mol^{-1}	1^{st}	2^{nd}	3^{rd}	4^{th}	5^{th}	6^{th}	7^{th}	8^{th}
Q	1090	2350	4610	6220	37800	47000	–	–
R	1251	2298	3822	5158	6542	9362	11018	33604

What is the likely formula of the compound that is formed when **Q** reacts with **R**?

A QR **B** Q_2R_3 **C** QR_4 **D** Q_4R

4 The following graph shows the first ionisation energy of 8 consecutive elements from **A** to **H** in the periodic table, with atomic numbers between 3 and 20.



Which of the following statements is **incorrect**?

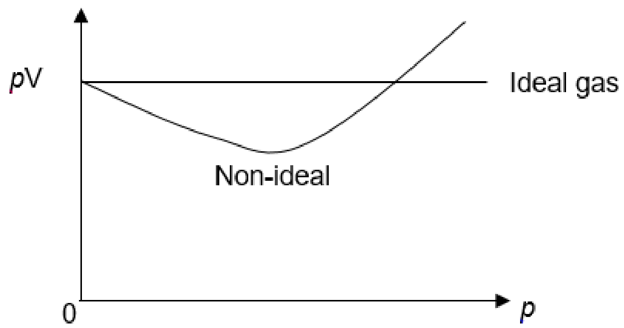
A The chloride of **E** has a giant ionic structure.

B The oxide of **F** has a giant molecular structure.

C The chloride of **D** conducts electricity in the molten state.

D Element **A** exists as discrete molecules at room temperature.

5 The value of pV is plotted against p for two gases, an ideal gas and a non-ideal gas, where p is the pressure and V is the volume of the gas.



Which gas shows the greatest deviation from ideality?

A ammonia

B methane

C ethene

D nitrogen

6 Iodine and phosphorus each forms a trifluoride.

What are the shapes of these two molecules?

	$\underline{IF_3}$	$\underline{PF_3}$
A	trigonal planar	trigonal pyramidal
B	T-shaped	trigonal pyramidal
C	trigonal pyramidal	trigonal planar
D	T-shaped	trigonal planar

7 Which of the following sets of compounds all contain the same type of forces of attraction between molecules of their own?

1 BF_3, CH_4, SF_6

2 $AlCl_3$, I_2, SiO_2

3 CO_2, NO_2, SiO_2

A 1 only

B 1 and 2 only

C 1 and 3 only

D 1, 2 and 3

8 Solid **X** has the following properties.

- It is insoluble in tetrachloromethane and water.
- It melts at 2730°C.
- It is a non-conductor of electricity.

Which of the following is a likely identity for **X**?

A sodium

B boron trifluoride

C boron nitride

D rubidium chloride

9 *The use of the Data Booklet is relevant to this question.*

A beaker containing 300 cm^3 of water, at room temperature and pressure, was heated above a spirit lamp containing propan-1-ol ($CH_3CH_2CH_2OH$). After heating for 5 minutes, the temperature of the water increased by 14.0 °C, while the mass of the lamp decreased by 0.52 g. Assuming that heat transfer is 80% efficient, what is the standard enthalpy change of combustion of propan-1-ol?

A −1621 kJ mol^{-1}

B −2026 kJ mol^{-1}

C −2532 kJ mol^{-1}

D +2532 kJ mol^{-1}

10 The enthalpy changes of the following reactions are:

$Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(s) + 3CO_2(g)$ $\Delta H = -26.8$ kJ mol^{-1}

$FeO(s) + CO(g) \rightarrow Fe(s) + CO_2(g)$ $\Delta H = -16.5$ kJ mol^{-1}

What is the enthalpy change for the reaction below?

$Fe_2O_3(s) + CO(g) \rightarrow 2FeO(s) + CO_2(g)$

A +6.2 kJ mol^{-1}

B –6.2 kJ mol^{-1}

C –10.3 kJ mol^{-1}

D –43.3 kJ mol^{-1}

11 The experimental values of some enthalpy change terms are given in the table below.

compound	NaBr(s)	KC*l*(s)
lattice energy / kJ mol^{-1}	−736	−698
sum of enthalpies of hydration of separate ions / kJ mol^{-1}	−754	−700

Which statement is correct?

A The difference in magnitude between experimental and theoretical lattice energies will be smaller for NaBr, compared to that for KC*l*.

B NaBr has more exothermic hydration energies than KC*l*, because the electronegativity difference between the ions in NaBr is greater than that in KC*l*.

C Upon dissolving, the ions in NaBr has stronger permanent dipole – permanent dipole attraction with water molecules, compared to those in KC*l* with water molecules.

D Enthalpy change of solution of solid NaBr is more exothermic than that of solid KC*l*.

12 Haber process is an important reaction which allows the industrial synthesis of ammonia. Finely divided iron is used as a catalyst for this reaction.

$$3H_2(g) + N_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H < 0$$

Which of the following statements about this system are correct?

1 The catalyst provides a surface for the reactant particles to react with each other.

2 The catalyst does not affect the equilibrium position.

3 For the same production rate of ammonia as before, the catalyst allows for a lower reaction temperature to be used.

A 1 only

B 1 and 2 only

C 1 and 3 only

D 1, 2 and 3

13 The rate law for the decomposition of a certain compound **Y** is rate = k [**Y**].

At 25 °C, k is found to be 5.37×10^{-3} s^{-1}. If the initial concentration of **Y** is 2.0×10^{-2} mol dm^{-3}, what is the concentration of **Y** after 387 seconds?

A 1.07×10^{-4} mol dm^{-3}

B 1.00×10^{-2} mol dm^{-3}

C 5.00×10^{-3} mol dm^{-3}

D 2.50×10^{-3} mol dm^{-3}

14 It has been suggested that the mechanism of the reaction between NO and O_2 involves the following steps.

$$2NO \rightarrow N_2O_2 \qquad \text{(fast)}$$
$$N_2O_2 + O_2 \rightarrow 2NO_2 \qquad \text{(slow)}$$

Which of the following statements are correct?

1 The stoichiometric equation is $2NO + O_2 \rightarrow 2NO_2$.

2 The rate equation is rate = k [N_2O_2] [O_2].

3 N_2O_2 is a catalyst.

A 1 only

B 1 and 2 only

C 1 and 3 only

D 1, 2 and 3

15 Solid copper and silver ions react in an equilibrium reaction as follows.

$$Cu(s) + 2Ag^{+}(aq) \rightleftharpoons Cu^{2+}(aq) + 2Ag(s)$$

Which of the following changes would shift the position of equilibrium to the right?

1 adding $AgNO_3$(aq)

2 adding water

3 using powdered copper instead of a lump of copper of equal mass

A 1, 2 and 3

B 1 and 2 only

C 2 and 3 only

D 1 only

16 Ammonium carbamate, $NH_4CO_2NH_2$, decomposes as follows:

$$NH_4CO_2NH_2(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$$

Which of the following statements about this system in equilibrium is correct?

A Increasing the volume of the system will shift the equilibrium position to the right.

B Increasing the pressure of the system will decrease the equilibrium constant, K_p.

C Addition of a catalyst will increase the rate of the production and hence increase the equilibrium constant, K_c.

D Reducing the concentration of CO_2(g) causes the equilibrium position to shift to the left.

17 An equilibrium was reached after 1 mole of solid I_2 was added to 50 cm^3 of 3 mol dm^{-3} KI.

$$I_2(s) + I^{-}(aq) \rightleftharpoons I_3^{-}(aq)$$

The reaction mixture was then rapidly cooled and filtered. Cold aqueous $AgNO_3$ was then added in excess to the cold filtrate. The mass of precipitate was 11.75 g.

Assuming that the equilibrium does not shift when temperature is low, calculate the value of K_c for the reaction.

A 2.22 **B** 2.00 **C** 0.15 **D** 0.11

18 The pK_b values of four weak monoacidic bases are shown in the table below.

base	**J**	**K**	**L**	**M**
pK_b	4.19	2.89	9.42	7.96

50.0 cm^3 of 0.1 mol dm^{-3} solutions of each base were prepared.

Which of the following statements is correct?

A The solutions need different amounts of acid to be completely neutralised.

B The pH of **J** will be higher than that of **L**.

C **K** is less basic than **M**.

D When water is added to 50.0 cm^3 of **L**, the pK_b value of **L** increases.

19 Which of the following could act as a buffer?

1 A mixture of 10 cm^3 of 0.10 mol dm^{-3} $NaHCO_3$(aq) and 15 cm^3 of 0.10 mol dm^{-3} Na_2CO_3(aq).

2 A mixture of 10 cm^3 of 0.10 mol dm^{-3} NaOH(aq) and 15 cm^3 of 0.10 mol dm^{-3} CH_3COOH(aq).

3 A mixture of 15 cm^3 of 0.10 mol dm^{-3} NH_3(aq) and 20 cm^3 of 0.10 mol dm^{-3} HC*l*(aq).

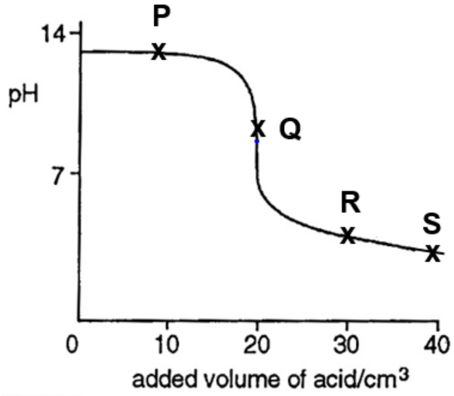
A 1 only

B 1 and 2 only

C 1 and 3 only

D 1, 2 and 3

20 The graph below shows the change in pH when 0.10 mol dm^{-3} ethanoic acid is gradually added to 10 cm^3 of 0.20 mol dm^{-3} sodium hydroxide.



Which of the following points refers to the maximum buffer capacity?

A point P

B point Q

C point R

D point S

21 A sparingly soluble aluminium salt dissociates in aqueous solution according to the equation below.

$$Al_2X_3(s) \rightleftharpoons 2Al^{3+}(aq) + 3X^{2-}(aq)$$

If the solubility product, K_{sp} of Al_2X_3 is S, what is the value of $[Al^{3+}(aq)]$ at equilibrium?

A $S^{\frac{1}{2}}$ **B** $(2/5\ S)^{\frac{1}{2}}$ **C** $S^{\frac{1}{5}}$ **D** $(8/27\ S)^{\frac{1}{5}}$

22 A solution containing 1.0×10^{-5} mol dm^{-3} of Ag^+ is slowly stirred into an equal volume of solution which contains Cl^- and Br^- ions, each with the same concentration of 2.0×10^{-5} mol dm^{-3}. The numerical values of the solubility products are:

species	K_{sp}
AgC*l*	1.8×10^{-10}
AgBr	3.3×10^{-13}

Which of the following statements is correct?

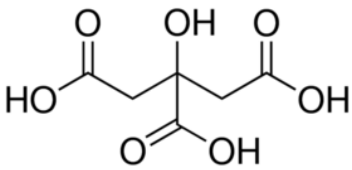
A No precipitate will be seen.

B Only AgBr precipitate will form.

C Only AgC*l* precipitate will form.

D A mixture of AgC*l* and AgBr precipitates will form.

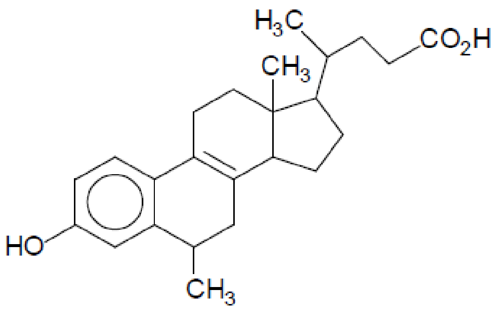
23 Citric acid is commonly found in citrus fruits such as lemon. It has the following structure:



How many sp^2 and sp^3 hybridised atoms are there in a molecule of citric acid?

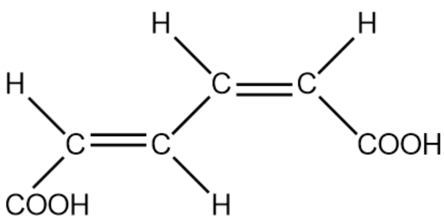
	sp^2 atoms	sp^3 atoms
A	3	4
B	3	7
C	6	4
D	6	7

24 How many enantiomers does this organic molecule have?



A 8 **B** 16 **C** 32 **D** 64

25 Muconic acid has the following structure:



Which of the following statements about muconic acid is true?

A It exists in three stereoisomeric forms.

B It contains two π bonds.

C There are four sp^2 hybridised carbon atoms.

D It will exhibit intra-molecular hydrogen bonding.

Index Number	Name and Form Class	Tutorial Class	Subject Tutor
__ __ __ __	1_____	1CH______	

ANGLO-CHINESE JUNIOR COLLEGE
DEPARTMENT OF CHEMISTRY
Promotional Examination

CHEMISTRY
Higher 2

9729

Answer Booklet

06 October 2020

Additional Materials: Multiple Choice Answer Sheet
Data Booklet
Question Booklet

2 hours

READ THESE INSTRUCTIONS FIRST
Write your index number and name on all the work you hand in.
Write in dark blue or black pen **except** on the Multiple Choice Answer Sheet.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.

Sections A, B and C
Answer **all** questions

Answer **all** questions in the spaces provided on the **Answer Booklet**. If additional space is required, you should use the pages at the end of this booklet. The question number must be clearly shown.

The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

You are reminded of the need for good English and clear presentation in your answers.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use		
	Question Number	**Marks**
Section A (25)	**1 – 25**	/25
Section B (25)	**1**	/6
	2	/6
	3	/13
Section C (25)	**1**	/15
	2	/7
	3	/3
Presentation of answers		
Total: 75		

Section B (25 marks)

Answer **all** the questions in this section on the spaces provided.

1 Nitryl fluoride, NO_2F is a colourless gas which acts as a strong oxidising agent and fluorinating agent.

It is known that NO_2F can be produced from nitrogen dioxide and fluorine according to the equation below.

$$2NO_2(g) + F_2(g) \longrightarrow 2NO_2F(g)$$

When NO_2 and excess F_2 were mixed together, $[NO_2]$ varied with time, as shown in the table below.

t / min	$[NO_2]$ / mol dm^{-3}	$[NO_2F]$ / mol dm^{-3}	$\frac{\Delta[NO_2F]}{\Delta t}$ / mol dm^{-3} min^{-1}
0.00	0.400	0.000	——
0.25	0.250	0.150	0.60
0.50	0.155	0.245	0.38
0.75	0.100		
1.00	0.060		
1.25	0.040		
1.50	0.025	0.375	0.06

(i) Explain why it is necessary to use excess fluorine in this experiment to confirm the order with respect to nitrogen dioxide.

……

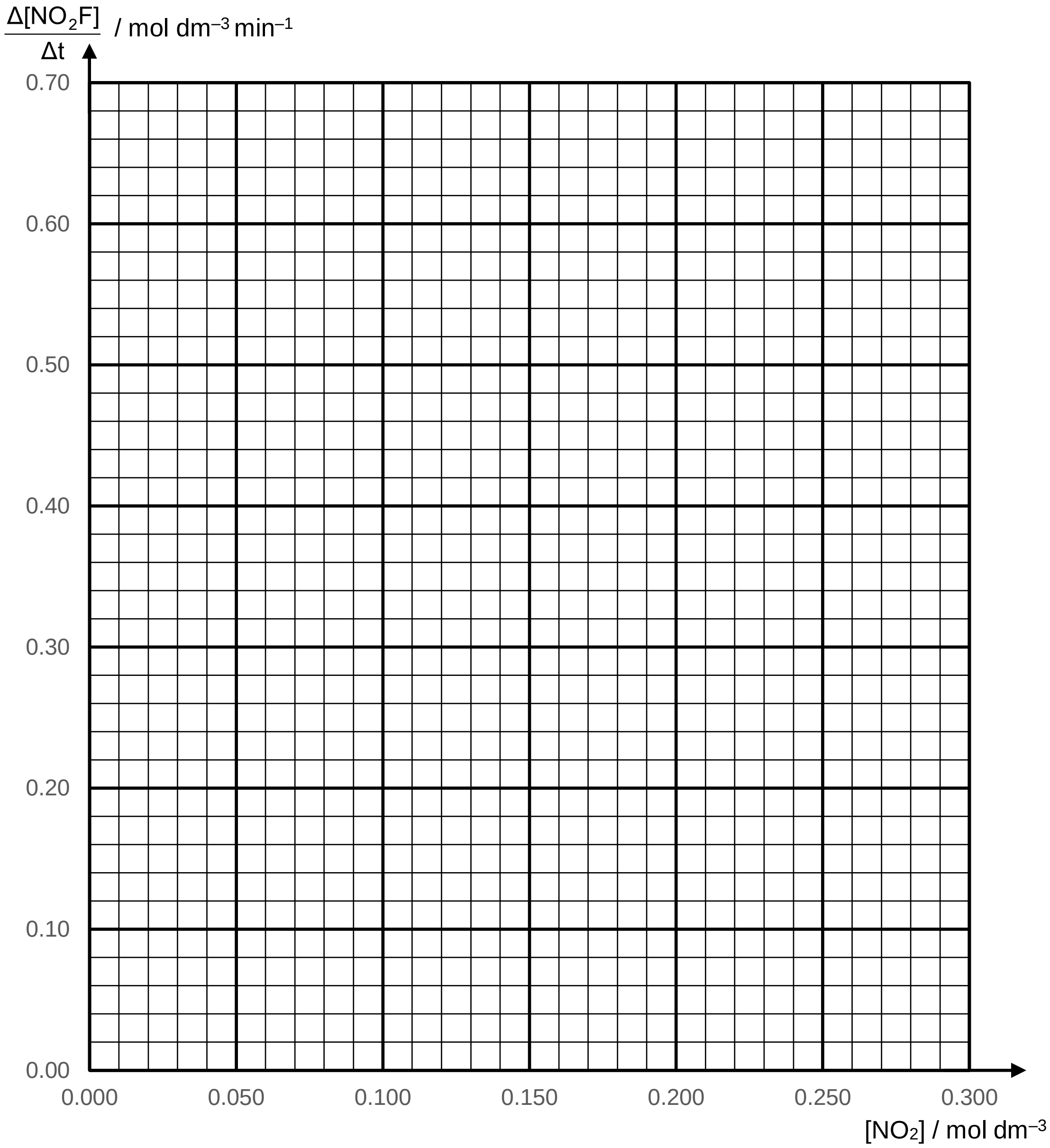
……

………………………………………………………………………………………………………[1]

(ii) The rate at a particular time can be approximated using rate = $\frac{\Delta[NO_2F]}{\Delta t}$.

Complete the table above by calculating $[NO_2F]$ produced (to 3 decimal places) and $\frac{\Delta[NO_2F]}{\Delta t}$ (to 2 decimal places) at the respective timings. [2]

(iii) Plot a graph of $\frac{\Delta[NO_2F]}{\Delta t}$ against $[NO_2]$ on the given grid and use it to deduce the order of reaction with respect to nitrogen dioxide.



…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

…………………………………………………………………………………………………..[3]

[Total: 6]

2 Lactic acid is a weak acid. It dissociates according to the equation shown.

$CH_3CH(OH)CO_2H(aq) \rightleftharpoons CH_3CH(OH)CO_2^-(aq) + H^+(aq)$ $K_a = 1.38 \times 10^{-4}$ mol dm^{-3}

(a) **(i)** Lactic acid is found in some anti-wrinkle cosmetics. A particular sample analysed contains 0.0150 mol dm^{-3} of lactic acid.

Calculate the pH of this sample. [2]

(ii) Buffered lactic acid is used in food confectionery.

A student prepared a similar buffered lactic acid solution for use in the laboratory. 25% of an original lactic acid solution was converted to its sodium salt with the addition of solid sodium hydroxide.

Calculate the pH of this buffered lactic acid solution. [1]

(iii) Write an equation to show how pH change is resisted when small amounts of hydrochloric acid is added to the buffered lactic acid.

……………………………………………………………………………………………...[1]

(b) Calcium lactate is used in medicine as a calcium supplement, and in the food industry and water treatment as a coagulant.

$$(CH_3CH(OH)CO_2)_2Ca(s) \rightleftharpoons 2CH_3CH(OH)CO_2^{-}(aq) + Ca^{2+}(aq)$$

Calcium lactate has a solubility of 0.0266 mol dm^{-3} at room conditions.

(i) Calculate the K_{sp} of calcium lactate. [1]

(ii) Calculate a value for ΔG at this temperature using $\Delta G = -RT\ln K$. Leave your answer in kJ mol^{-1}. [1]

[Total: 6]

3 Hennig Brand discovered phosphorus in 1669. It is the first element to be discovered that was not known since ancient times and is essential for life. Not surprisingly, Brand's original procedure involved heating urine until a white substance was obtained. Ammonium sodium hydrogen phosphate, $(NH_4)NaHPO_4$, is the main compound in urine which produced the white substance, sodium metaphosphate, $NaPO_3$, on thermal decomposition. In the process, two gases are given off.

(a) **(i)** Write a balanced equation for the thermal decomposition of $(NH_4)NaHPO_4$.

……………………………………………………………………………………………..[1]

(ii) Given a pure sample of solid $(NH_4)NaHPO_4$, briefly outline the experimental method to be performed to ensure that the thermal decomposition of $(NH_4)NaHPO_4$ is complete.
You do not need to specify quantities used.

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

……………………………………………………………………………………………..[2]

Phosphorus derives its name from Greek words which mean *'light-bringer'*. In later works, the white substance was heated with sand and graphite to liberate phosphorus which glowed in the dark.

$$4NaPO_3 + 2SiO_2 + 10C \rightarrow 2Na_2SiO_3 + 10CO + P_4$$

(b) **(i)** Use the information above and concepts from Atomic Structure to suggest how light energy is released from phosphorus molecules.

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

……………………………………………………………………………………………..[2]

(ii) Explain, in terms of structure and bonding, why the melting point of silicon dioxide, SiO_2, is higher than carbon monoxide, CO.

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

……………………………………………………………………………………………..[2]

Phosphorus exists in various allotropic forms and P_4 forms a vapour which dissociates into diatomic P_2 molecules at high temperatures.

$$P_4(g) \rightleftharpoons 2P_2(g)$$

When 0.883 g of P_4 was heated to 1200 ^oC in a 10.0 dm^3 vessel and left to equilibrate, the total pressure at equilibrium measured was 0.124 atm.

(c) **(i)** Calculate the initial pressure of P_4 in atm, assuming no dissociation has taken place yet. [2]

(ii) Hence, calculate the partial pressures of P_4 and P_2 at equilibrium. [1]
*If you were unable to obtain a value in **(c)(i)**, use 0.100 atm as the initial pressure of P_4, although this is not the correct value.*

(iii) Write an expression for K_p and include its unit. [2]

(iv) Calculate a value for K_p. [1]

[Total:13]

Section C (25 marks)

Answer **all** the questions in this section on the spaces provided.

1 Sulfur and phosphorus are both Period 3 elements.

(a) Generally, ionisation energies increase across a period.

(i) Explain why ionisation energies increase generally across a period. [2]

(ii) Explain why phosphorus has the higher first ionisation energy than sulfur instead. [1]

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....

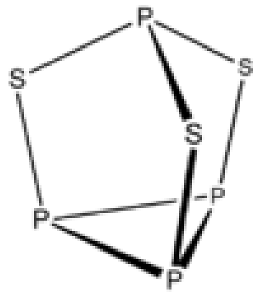
....

....

....

....

(b) Sulfur forms a wide range of sulfides with phosphorus. One of the most well-known examples is the three-fold symmetric P_4S_3 used in strike-anywhere matches.



(i) Suggest the bond angle about each sulfur atom in P_4S_3. [1]

(ii) The bond energies of the P—P and P—S bonds are 197 kJ mol^{-1} and 270 kJ mol^{-1} respectively.

Given the enthalpy change for the following transformation

$$P_4S_3(g) \longrightarrow P_4(g) + 3S(g) \qquad \Delta H = +1029 \text{ kJ mol}^{-1},$$

show that there are six P—P bonds in P_4. [2]

(iii) Given that the four phosphorus atoms are identical in P_4, draw the displayed formula of P_4 and state the shape about each atom. [2]

(iv) P_4O_3 has a structure that is similar to P_4S_3.
Explain why the O—P—O bond angle is smaller than the S—P—S bond angle. [2]

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

(c) The reaction scheme below shows the reactions of some manganese compounds.

$$MnO_2 \xrightarrow[\textbf{Step I}]{O_2/\ KOH} MnO_4^{2-} \xrightarrow[\textbf{Step II}]{\text{dilute acid}} MnO_2 + \text{purple solution of } KMnO_4$$

In **step II**, reaction of MnO_4^{2-} with dilute acid produced 0.174 g of MnO_2 and a purple solution of $KMnO_4$. The $KMnO_4$ solution was found to react completely with 26.70 cm^3 of 0.75 mol dm^{-3} iron(II) sulfate in an acidic medium.

(i) State the type of redox reaction in **step II**. [1]

(ii) Given that the reaction between $KMnO_4$ and iron(II) sulfate is given as:

$$5Fe^{2+} + MnO_4^- + 8H^+ \longrightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$$

Show that MnO_2 and $KMnO_4$ are formed in the mole ratio 1:2 in **step II**. [2]

(iii) Construct the balanced ionic equation for the reaction that occurred in **step II**. [2]

[Total: 15]

2 Nitrogen is one of the most abundant elements on earth. It is present in many molecules that find applications in chemistry and biology, such as ammonia, haemoglobin and amino acids.

(a) During the combustion of ammonia, it reacts to form nitrogen gas and water vapour as seen in the equation below:

$$4NH_3(g) + 3O_2(g) \longrightarrow 2N_2(g) + 6H_2O(g) \quad - \quad \text{reaction } \mathbf{(1)}$$

(i) State the role of ammonia in reaction **(1)**. [1]

The standard entropies, $S^\ominus$, of the above species are tabulated.

	$S^\ominus$ / J mol^{-1} K^{-1}
$NH_3(g)$	192.8
$O_2(g)$	205.2
$N_2(g)$	191.6
$H_2O(g)$	188.8

(ii) The standard change of entropy of a reaction is the difference between the sum of the standard entropies of the products and the sum of the standard entropies of the reactants.

Calculate the standard entropy change of reaction **(1)**. [1]

(iii) Account for the sign in your answer to **(ii)**. [1]

(iv) The standard enthalpies of formation of ammonia and $H_2O(g)$ are –45.9 kJ mol^{-1} and –241.8 kJ mol^{-1} respectively.

Calculate the standard enthalpy change of reaction **(1)**. [1]

(v) Hence, calculate the $\Delta G^\ominus$ of reaction **(1)**. [1]

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

………………………………………………………………………………………….............

………………………………………………………………………………………….............

………………………………………………………………………………………….............

………………………………………………………………………………………….............

………………………………………………………………………………………….............

………………………………………………………………………………………….............

…………………………………………………………………………………………………

………………………………………………………………………………………….............

(b) $4NH_3(g) + 3O_2(g) \longrightarrow 2N_2(g) + 6H_2O(l)$ - reaction **(2)**

The standard enthalpy change of reaction **(2)** is –1531.2 kJ mol^{-1}.

With the aid of an energy level diagram, find the standard enthalpy change of vaporisation of water. [2]

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

[Total: 7]

3 Pentanoic acid is a carboxylic acid with molecular formula $C_5H_{10}O_2$.

Take note that your answers to **(a)**, **(b)** and **(c)** should **NOT** be the same compound.

(a) Draw the **stereochemical** formula of a constitutional isomer of $C_5H_{10}O_2$ which has chiral carbons but is achiral. [1]

Like other low-molecular-weight carboxylic acids, pentanoic acid has an unpleasant odour. In contrast, the esters of pentanoic acid tend to have pleasant odours and are used widely in perfumes and cosmetics.

(b) Draw the **skeletal** formula of a chiral ester which is a constitutional isomer of $C_5H_{10}O_2$. [1]

(c) Draw the **skeletal** formula of a constitutional isomer of $C_5H_{10}O_2$ which does not show enantiomerism but shows cis-trans isomerism. [1]

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

[Total: 3]

Additional answer space

If you use the following pages to complete the answer to any question, the question number must be clearly shown.